JOURNAL OF SPACECRAFT AND ROCKETS
Vol. 55, No. 4, July–August 2018

Check for updates

Engineering Notes

Extension of the Simple Analytical Model for Waverider Design

Dan Igra*
Aerodynamics Group, 31021 Rafael, Israel

DOI: 10.2514/1.A34089

Downloaded by rafael library on July 27, 2018 | http://arc.aiaa.org | DOI: 10.2514/1.A34089

Nomenclature

A, B, C	=	power-law scaling parameters
D	=	drag
h	=	height
L	=	lift
l	=	length
M	=	Mach number
M_y	=	pitch moment
m, n	=	power-law exponents
w	=	width
x, y, z	=	coordinates
α	=	angle of attack
β	=	oblique shock inclination angle
γ	=	specific heat ratio
δ	=	leading-edge inclination angle
θ	=	centerline wedge angle

Subscripts

base	=	value for base surface
cp	=	center of area
l	=	value of lower surfaces
ref	=	pitch moment reference location
u	=	value of upper surface
∞	=	freestream value

I. Introduction

THE present paper describes an extension of previous work of Starkey and Lewis [1] in which a simple analytical model for hypersonic waverider design was discussed. This method takes a power-law waverider and treats it as a series of wedges in which the shock or expansion waves are solved analytically. Some viscous effects are added as well. Several examples are solved in [1] and are compared with results based on more complex codes for waverider design; good agreement is found between the two. Examples of possible waverider geometries designed with this method are shown in Figs. 1 and 2.

Here, two new aspects shall be addressed. The first is the calculation of the pitch moment. The second is the estimation of the waverider's base pressure. In the previous work of Starkey and Lewis [1], no equation for calculating the pitch moment was given. Their main concern was the lift and drag forces acting on the waverider. These are obviously the most important parameters in a waverider design. However, the addition of a pitch moment is of great importance because, along with the lift and drag, it can be used for estimating the waverider trajectory. This offers significant advantage when employing a simple analytic model for waverider design combined with trajectory code to study whether the proposed design meets the required range and trajectory. Most waverider's design codes do not estimate the pitch moment, and to the best of my knowledge, this is the first time that such addition is introduced to the simple analytic model for the waverider design of [1]. The second addition is an empirical calculation for the base pressure. In [2], there was a discussion about the base pressure, and the recommendation was that it would be either zero or equal to the freestream pressure. In the current study, when dealing with a gliding nonpowered waverider, both of these assumptions are not correct. Although the base pressure decreases with increase in the flight Mach number, it will be larger than zero and lower than the freestream pressure. Here, an empirical correlation obtained from [3] shall be employed.

To demonstrate how these new capabilities extend the simple analytic model for waverider design, a comparison with experimental and numerical results of a waverider shall be presented subsequently. Good agreement was obtained for the drag, lift, and pitch moment for all Mach numbers tested within the limits of the assumptions employed.

Next, a description of how to calculate the pitch moment shall be presented, followed by an empirical correlation for the base pressure. In Sec. IV, available wind-tunnel results obtained for a waverider will be compared with the results of the extended simple analytic model for waverider design and with computational fluid dynamics (CFD) results.

II. Extended Simple Analytic Model for Waverider Design: Pitch Moment Calculation

By employing the same assumptions as in [1] and the power-law derived for the waverider geometry, it is possible to obtain an analytic expression for the pitch moment. The assumptions for calculating the pitch moment are as follows.

1) The shock or expansion waves begin at the leading edge of the waverider and continue at the same initial deflection angle. This results in a uniform pressure along the surface.

2) The flow is considered to be an ideal gas with constant γ.

Based on these assumptions, the pitch moment can be evaluated employing the following method. First, locate the area center of the planform (see in Fig. 1) and the base of the waverider. Based on the assumption that the pressure is evenly distributed along the surfaces, the force acting on this surface will be at the center of the area. By solving two equations, it is possible to obtain the centers in the x direction for the planform and the z direction for the base.

For the simpler case, called the constant wedge angle waverider in [1], whose geometry is shown in Fig. 1 (taken from [1]), after some manipulation, the center of the area in the x and z directions is obtained from the following expressions:

$$x_{\text{cp}} = \frac{\int_0^l xAx^n \,\mathrm{d}x}{\int_0^l Ax^n \,\mathrm{d}x} = \frac{n+1}{n+2} l \tag{1}$$

$$z_{\text{cp}} = \frac{\int_0^{Al^n} \int_{z_u}^{z_l} z \,\mathrm{d}z \,\mathrm{d}y}{\int_0^{Al^n} \int_{z_u}^{z_l} \mathrm{d}z \,\mathrm{d}y} = \frac{l(n\tan(\beta) + \tan(\theta))}{(n+2)} \tag{2}$$

For the more complex variable wedge waverider, whose geometry is shown in Fig. 2, the center of the area in the x direction is the same

Received 19 September 2017; revision received 14 December 2017; accepted for publication 19 December 2017; published online 12 March 2018. Copyright © 2017 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the ISSN 0022-4650 (print) or 1533-6794 (online) to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

*Research Associate.

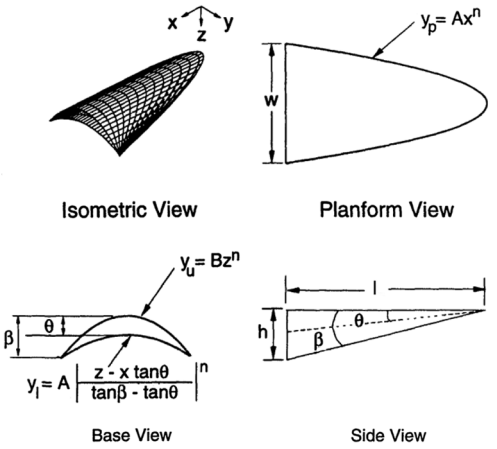


Fig. 1 Schematic description of a constant wedge angle waverider (taken from [1]).

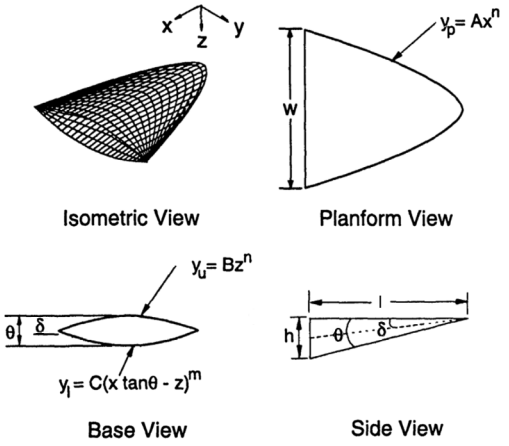


Fig. 2 Schematic description of a variable wedge angle waverider (taken from [1]).

as in the previous constant wedge case. However, the center of the area in the z direction is given in the following equation:

$$z_{\text{cp}} = \frac{\int_0^{Al^n}\int_{z_u}^{z_l} z\,\mathrm{d}z\,\mathrm{d}y}{\int_0^{Al^n}\int_{z_u}^{z_l} \mathrm{d}z\,\mathrm{d}y} \tag{3}$$

resulting in

$$z_{\text{cp}} = \frac{l(m-n)\tan^2(\delta)/[(m+2)(n+2)] + l\tan(\theta)(\tan(\theta) + m\tan(\delta))/[(m+2)(m+1)]}{\tan(\theta)/(m+1) + \tan(\delta)(m-n)/[(m+1)(n+1)]} \tag{4}$$

Once the center locations are calculated, the pitch moment can be calculated by the following equation:

$$M_y = (x_{\text{cp}} - x_{\text{ref}}) * L + (z_{\text{cp}} - z_{\text{ref}}) * D \tag{5}$$

where x_{ref} and z_{ref} are the reference points where the pitch moment is acting on.

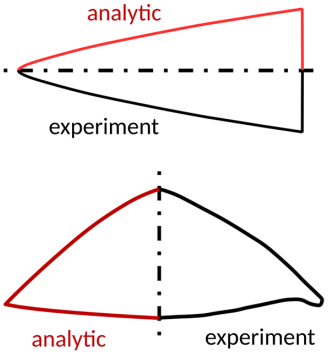


Fig. 3 Comparison between planform and base views of the experimental and analytical power-law-based waverider geometry.

III. Extended Simple Analytic Model for Waverider Design: Base Pressure Correlation

The base pressure correlation was taken from [3]; it is the following equation based on the freestream Mach number:

$$c_{p_{\text{base}}} = \frac{2}{\gamma M_\infty^2}\left[\left(\frac{2}{\gamma+1}\right)^{1.4}\left(\frac{1}{M_\infty}\right)^{2.8}\left(\frac{2\gamma M_\infty^2 - (\gamma-1)}{\gamma+1}\right) - 1\right] \tag{6}$$

IV. Results and Discussion

To validate the addition of these new features to the simple analytic model for waverider design, a comparison with results obtained from [4,5] while testing a waverider model in a hypersonic wind tunnel is presented subsequently. The shape of the waverider computed in the present, extended simple analytical model was slightly modified; it is presented by the power-law model. A comparison between the two geometries is presented in Fig. 3 for both planform and base views.

As can be seen, there are some differences between them. The wind-tunnel model has a round leading edge, whereas that of the simple analytic model for waverider design has a sharp leading edge. The geometrical parameters used in the present, simple analytic model for waverider design are $l = 39$ in., $w = 16$ in., $\theta = 10$ deg, $\delta = 9$ deg, $m = 0.6513$, and $n = 0.7215$. (For definition of these parameters, see Fig. 2.)

As in [4,5], the reference area is the planform area of 0.242 m^2 for a full configuration. The reference location for the pitch moment is located at the tip of the nose at $x = y = z = 0$ m, the same as in [4,5]. This waverider was designed for a Mach number of 14. It was designed using the MAXWARP code with corrections for round leading edges. Experiments were carried out in the Arnold Engineering Development Complex (AEDC) tunnel 9, and obtained results are reported in [4,5]. A photo of the wind-tunnel model is shown in Fig. 4 in [4]. In the reported experiments, the total pressure and temperatures were 9797 psi and 1707 K; these are summarized in Table 1. Results obtained for the drag, lift, and the pitch moment coefficients, as well as for L/D for this case, are shown in Fig. 4. For comparison, CFD results are added; these results are based on the geometry of the simple analytic model proposed for waverider design with the addition of very small rounded leading edges. Overall, there appears to be good agreement between experimental, numerical, and

Downloaded by rafael library on July 27, 2018 | http://arc.aiaa.org | DOI: 10.2514/1.A34089

Table 1 Flow conditions for AEDC tunnel 9 experiment taken from [4]

Experiment number	Mach number	Total pressure	Total temperature
2389	14	9797 psi	1707 K

Table 2 L/D at zero angle of attack deduced using different methods

Data source	L/D
Experiment [5]	3.7
MAXWARP sharp leading edges [5]	4.61
MAXWARP round leading edges [5]	4.07
Analytic $p_{\text{base}} = f(M)$	4.31
Analytic $p_{\text{base}} = 0$	4.278
Analytic $p_{\text{base}} = p_\infty$	4.59
CFD	3.94

analytical results. This is quite surprising considering that this is a simple analytic code with a slightly modified geometry.

First, let us examine the results for L/D. Here, two additional data points are present; these were computed with the MAXWARP code and obtained from [5]. They are L/D as computed by MAXWARP for sharp leading edge and round leading edge, the same as those of the experimental model. A summary of these results is presented in Table 2. The round leading edges are vital for heat transfer, but they cause high-pressure air to escape from the lower surface of the waverider toward its upper surface, where a lower pressure exists, as shown in Fig. 5. As a result, the lift and L/D decrease. For this reason, the current results for the simple analytic model for waverider design are higher than those obtained in the experiment. In addition, at negative angles of attack (-5 deg and lower), both the experimental and the CFD results for the lift coefficient change their slope angle. This is due to the shock wave under the lower surface of the waverider that does not maintain a constant angle. Computing this accurately is beyond the scope of a simple design code. In spite of that, the results for both drag and pitch moment coefficients have good agreement with both experiments and CFD findings. This is a confirmation that a simple analytical model for waverider design can obtain good agreement with real experimental and CFD findings, making it a viable tool for initial design of waverider configurations.

In Table 2, L/D for zero angle of attack for the case presented in Fig. 4 are shown. In addition, the effect of the base pressure on L/D is also included when computed with the simple analytic model for waverider design. The base pressure was also computed for cases where it was set to either zero or the freestream value. It is interesting to see that, when the base pressure is equal to the freestream pressure, the L/D value computed by the simple analytic model for waverider

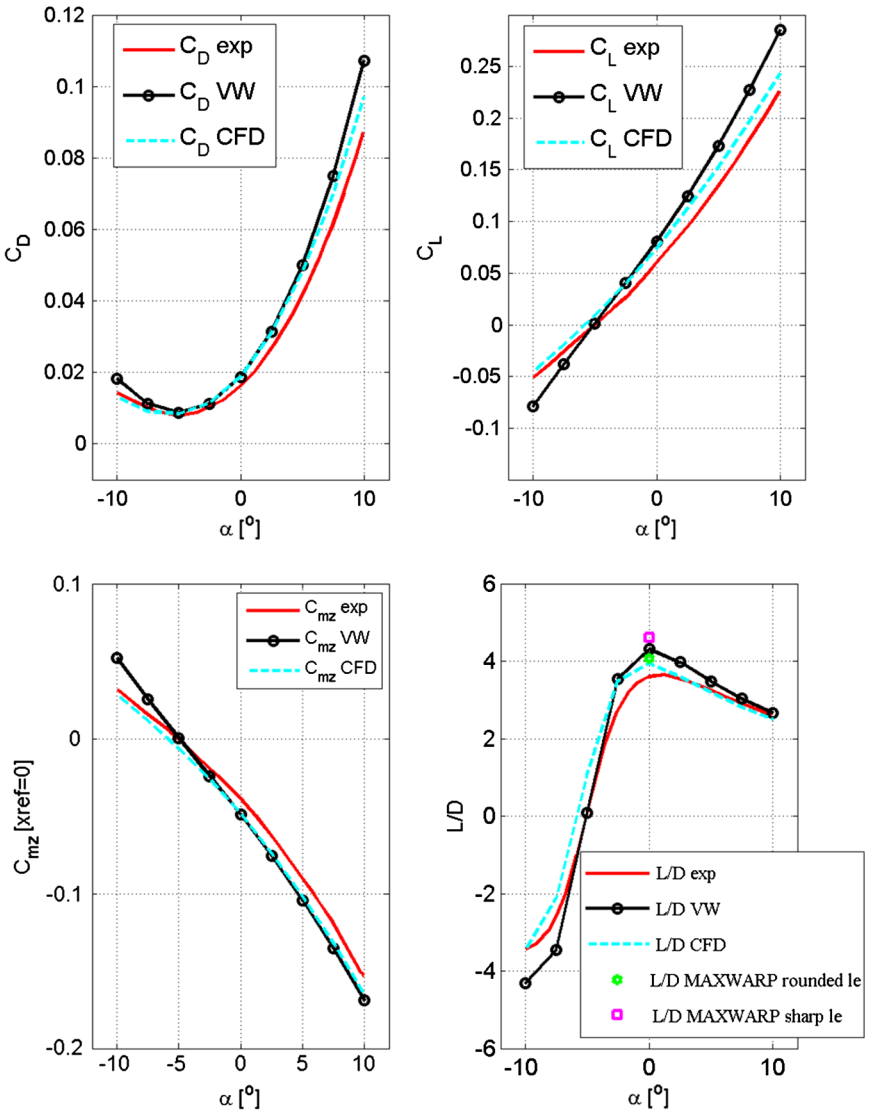


Fig. 4 Comparison between the aerodynamic coefficients obtained by analytical, experimental, and CFD results, for waverider at $M = 14$.

Downloaded by rafael library on July 27, 2018 | http://arc.aiaa.org | DOI: 10.2514/1.A34089

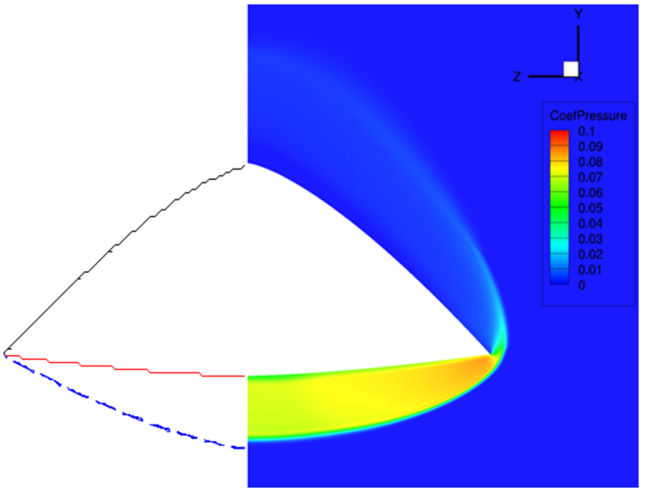


Fig. 5 Comparison between analytically obtained shock wave location and the pressure coefficient computed via CFD, at the base of the waverider, at zero angle of attack.

design is almost the same as that of the MAXWARP with sharp leading edges. The results for zero base pressure and that of the base pressure equation based on freestream Mach number taken from [3] are close. It can be clearly seen that, in spite of the high-Mach-number flow, the base pressure still has a significant influence on L/D, with a difference of about 6% when computed with the simple analytic model for waverider design. This is especially true for a nonpowered waverider glider or in a case when a glider model is tested in a wind tunnel. Thus, employing the freestream pressure as the base pressure for these cases will result in an overoptimistic result.

V. Conclusions

The suggested improvements to the simple analytical model proposed for waverider design demonstrated the ability of this simple variable wedge method to predict accurately the aerodynamic properties. This method has been extended in two ways; one is the calculation of the pitch moment, and the other an estimation of the base pressure. Employing these additions resulted in very good agreement with both experimental and CFD results.

References

[1] Starkey, R. P., and Lewis, M. J., "Simple Analytical Model for Parametric Studies of Hypersonic Waveriders," *Journal of Spacecraft and Rockets*, Vol. 36, No. 4, 1999, pp. 516–523. doi:10.2514/3.27194

[2] Starkey, R. P., and Lewis, M. J., "Analytical Off-Design Lift-to-Drag-Ratio Analysis for Hypersonic Waveriders," *Journal of Spacecraft and Rockets*, Vol. 37, No. 5, 2000, pp. 684–691. doi:10.2514/2.3618

[3] Guidi, A., Chu, Q. P., Mulder, J. A., and Buursink, J., "Implementation of an Aerodynamic Toolbox in a Reentry Flight Simulator," *Journal of Spacecraft and Rockets*, Vol. 40, No. 1, 2003, pp. 138–141. doi:10.2514/2.3929

[4] Norris, J. D., "Mach 8 High Reynolds Number Static Stability Capability Extension Using a Hypersonic Waverider at AEDC Tunnel 9," *25th AIAA Aerodynamic Measurement Technology and Ground Testing Conference*, AIAA Paper 2006-2815, June 2006. doi:10.2514/6.2006-2815

[5] Gillum, M. J., and Lewis, M. J., "Experimental Results on a Mach 14 Waverider with Blunt Leading Edges," *Journal of Aircraft*, Vol. 34, No. 3, 1997, pp. 296–303. doi:10.2514/2.2198

G. E. Palmer
Associate Editor

Downloaded by rafael library on July 27, 2018 | http://arc.aiaa.org | DOI: 10.2514/1.A34089